Problem Statement

Title: Sort an Array of Integers in Ascending Order Using a Function (No For-Each Loop)

Write a function called `sortArray` that takes an integer array as input, sorts the elements in ascending order using a basic sorting algorithm (without using built-in sort functions), and returns the sorted array. In your program, print both the original and the sorted array using only traditional for loops (not for-each).

Sample Input

text

```
Input array: [5, 3, 8, 2, 1]
```

Expected Output

text

```
Input array: 5 3 8 2 1
Sorted array: 1 2 3 5 8
```

Java Solution (with Only Normal for Loops)

java

```java
public class ArraySorter {

    // Function to sort the array using Bubble Sort
    public static int[] sortArray(int[] arr) {
        int n = arr.length;
        // Implement Bubble Sort
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
```

```java
                    // Swap arr[j] and arr[j+1]
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
        return arr;
    }

    public static void main(String[] args) {
        int[] inputArray = {5, 3, 8, 2, 1};

        System.out.print("Input array: ");
        for (int i = 0; i < inputArray.length; i++) {
            System.out.print(inputArray[i] + " ");
        }

        // Call the sortArray function
        int[] sortedArray = sortArray(inputArray);

        System.out.print("\nSorted array: ");
        for (int i = 0; i < sortedArray.length; i++) {
            System.out.print(sortedArray[i] + " ");
        }
    }
}
```